

AETNA HEALTHCARE 360 – DATA ANALYSIS, DAX & DASHBOARD PROJECT

Excel Analysis + Power BI DAX + Interactive Dashboard Documentation

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Course Name : Program In AI Driven Data Analysts

Batch No : FNB16

PROJECT TITLE: Aetna Healthcare 360 – Integrated Healthcare Analytics and Revenue Cycle Dashboard

Project Overview

This project analyses an integrated healthcare dataset using Excel and Power BI. The Excel workbook contains cleaned/master tables, calculated fields and PivotTable-based summaries, while the PBIX file contains a multi-page interactive Power BI report with DAX measures, cards, charts, tables and slicers

Health Track 360 is a data-driven health analytics project focused on transforming raw health data into clear, actionable insights. The process involved data cleaning and transformation in Excel using advanced functions, descriptive statistics, and pivot tables. The cleaned data was then modelled in Power BI with table relationships and visualized through an interactive dashboard.

This end-to-end workflow helps healthcare stakeholders to track key health metrics, identify trends and risk areas, and make informed, data-backed decisions.

Project Objective

The main objective is to convert patient, encounter, clinical, procedure, prescription, provider, vitals and insurance-claim data into actionable insights covering patient demographics, clinical activity, service revenue, pharmacy performance, provider/facility performance and revenue-cycle efficiency.

The primary objectives of this project are:

Data Cleaning & Preparation: Preprocess raw health data in Excel to ensure accuracy, consistency, and completeness.

Advanced Excel Analysis: Apply Pivot Tables, XLOOKUP formulas, and descriptive statistics such as mean, median, mode, and standard deviation to derive insights.

Data Modelling: Create relationships between tables in Power BI to build a robust data model for analysis.

Interactive Visualization: Develop a Power BI dashboard with slicers, KPIs, charts, and filters to present health metrics clearly.

Decision Support: Enable stakeholders to monitor health indicators and identify patterns for better decision making.

Technical Skill Demonstration: Showcase end-to-end skills in Excel data wrangling, statistical analysis, and Power BI dashboard/reporting.

Data Sources

- **Source Description and Timeline:** Healthcare dataset obtained from XExcelX website. The data covers the period 2022-2026 and includes patient demographics, health indicators, hospital records, and treatment outcomes.
 - **Link:** [The Ultimate Healthcare Dataset Generator for Excel & Power BI | Excelx.com](#)
 - **Domain:** Healthcare Analytics / Public Health / Hospital Management
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3. Problem Statement

- Analyses patient demographics and routine-care patterns to understand the population served.
 - Evaluate encounter volume and charge revenue by encounter type, CPT/service and facility.
 - Identify major diagnoses and clinical patterns contributing to healthcare activity and charges.
 - Measure provider and specialty performance to identify high revenue providers and staffing patterns.
 - Assess prescription volume, medication revenue and pharmacy performance.
 - Evaluate insurance billing, collections and outstanding balances to support revenue-cycle decisions.
 - Monitor vital-sign and risk indicators for population-health oriented analysis.
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4. Attribute (Column /Features) Details:

Attribute Name	Data Type	Description
Patient ID	Integer / String (Text Categorical)	Unique identifier for each patient. Format: PAT-1001, PAT-1002, etc. Used for tracking and linking records.
Full Name	String (Text)	Full name of the patient. Used for identification and demographic analysis.
Date of Birth	Numeric (Integer)	Patient's date of birth in YYYY-MM-DD format. Used to calculate age and analysed age-group health trends.
Gender	Categorical	Gender of the patient. Values: Male, Female, Non-binary. Used for demographic and health disparity analysis.
Blood Type	Categorical	Patient's blood group. Values: A+, A-, B+, B-, AB+, AB-, O+, O-. Used for medical compatibility and health studies.
Age / Age Group	Integer / Categorical	Patient age and derived demographic band Age: Patient age in years at time of encounter. Eg: 45. Age Group: Derived band from Age for analysis & privacy. Eg: 0-17, 18-35, 36-50, 51-65, 65+.

EncounterID	String - Primary Key	Unique healthcare encounter identifier. Unique ID for ONE visit. One Patient can have 10 EncounterIDs. Used to join Diagnosis, Procedure and Billing tables.
Encounter Date	Date - YYYY-MM-DD	Date of healthcare encounter. Actual date of service. Used for time-series, seasonality, monthly revenue, and patient visit frequency. Check for future dates as data error.
Encounter Type	Categorical - 4 values	Setting of care. These drives cost heavily: Emergency: ER - High cost, unplanned Hospital Admission: Inpatient - Highest cost Office Visit: OPD / Clinic - Routine Telehealth: Virtual - Lowest cost
ProviderID / Provider Name	String	Healthcare provider identifier and name. This is used to measure provider performance and high-revenue specialties as per your objective.
FacilityID	String	Hospital/clinic location identifier. Where the care happened, Used for hospital-wise revenue, location-wise utilization, and staffing pattern analysis.
ICD Code / Diagnosis Description	String	Diagnosis classification and description, used for population health, risk scoring, and comorbidity analysis. One encounter can have multiple ICDs.
CPT_Code	Integer	Procedure/service billing code, this decides the charge. Used to assess service volume and procedure cost.
Charge Amount	Decimal	Charge associated with an encounter/service. Total money billed for one full encounter / visit. This is GROSS charge before insurance discount.
PrescriptionID / Medication ID	String	Prescription and medication identifiers, used to link to Pharmacy master table to get drug name, dosage. Used for your goal: "Assess prescription volume and pharmacy performance".
Charges	Decimal	Prescription charge amount, this one is specifically cost of the medicine only. Eg: \$45.50 for a drug. Separate it from Encounter Charge Amount. Rename it to Prescription Charge for clarity.
ClaimID / ChargeID	String	Insurance claim and linked charge identifiers, ClaimID groups multiple ChargeIDs together for one insurance claim. Used to track if claim was accepted, denied, paid. This is the backbone of Revenue Cycle Management.

Amount Billed / Amount Paid / Remaining	Decimal	Revenue-cycle billing, payment and outstanding balance. Use this to find collection efficiency, bad debt, and pending AR
SystolicBP / DiastolicBP / HeartRate	Numeric	Recorded vital-sign measures. Used for analysing and monitor vital-sign and risk indicators for population-health analysis

5. Tools & Technologies

- **Excel:** Data cleaning, helper columns, categorisation, PivotTables, summaries and validation checks.
- **Power BI:** Data modelling, DAX measures, cards, charts, slicers, drill-down and consolidated reporting.

6. Data Pre-Processing (Excel / Power Query)

Tasks Performed:

- Removed/ignored blank analysis rows by validating primary-key fields before analysis.
- Standardised and used identifier fields such as PatientID, EncounterID, ProviderID, FacilityID, ChargeID and ClaimID.
- Created/used derived categories including Age Group, charge status/tier, service type, claim-size category, medication category and clinical risk categories.
- Used PivotTables for demographic, revenue, diagnosis, procedure, pharmacy, insurance and provider summaries.
- Validated totals across linked summaries; for example, 5,614 charge records are represented in the Excel CPT revenue/volume summary.
- Logical fact/dimension structure is possible:
Encounters/Charges/Claims/Prescriptions/Procedures/Vitals/Diagnoses as activity facts, with Patients, Providers, Facilities, Medications and Pharmacies as dimensions.

	A	B	C	D	E	F	G	H	I	J
1	PatientID	First Name	Last Name	DateOf	Age	Age Group	Gender	BloodType	Checkup Routine	
2	PAT-1001	Jennifer	Brown	1973-09-2	52	Adult	Female	A-	Regular	
3	PAT-1002	Mary	Johnson	1996-02-2	30	young	Male	AB+	Regular	
4	PAT-1003	John	Smith	1967-04-1	59	Adult	Non-binary	O+	Regular	
5	PAT-1004	Aarav	Singh	1970-11-0	55	Adult	Female	AB+	Regular	
6	PAT-1005	Priya	Brown	2007-11-2	18	young	Male	B-	Regular	
7	PAT-1006	James	Jones	1994-09-2	31	Adult	Male	B+	Regular	
8	PAT-1007	Jennifer	Brown	2002-06-2	24	young	Male	AB-	Regular	
9	PAT-1008	Mary	Singh	2001-02-2	25	young	Non-binary	A+	Regular	
10	PAT-1009	John	Jones	1991-07-0	35	Adult	Female	AB+	Regular	
11	PAT-1010	Aarav	Johnson	1987-07-0	39	Adult	Female	B-	Regular	
12	PAT-1011	Mary	Brown	1942-07-2	83	Senior Citizer	Non-binary	O-	Checkup Needed	
13	PAT-1012	James	Garcia	2006-06-1	20	young	Female	AB-	Regular	
14	PAT-1013	John	Williams	1963-06-1	63	Senior Citizer	Female	AB-	Checkup Needed	
15	PAT-1014	Patricia	Johnson	1963-08-2	62	Senior Citizer	Female	O+	Checkup Needed	
16	PAT-1015	Michael	Williams	1944-05-0	82	Senior Citizer	Male	O-	Checkup Needed	
17	PAT-1016	James	Singh	1996-11-0	29	young	Female	B-	Regular	
18	PAT-1017	James	Patel	1979-01-2	47	Adult	Male	AB-	Regular	
19	PAT-1018	Robert	Smith	1967-03-3	59	Adult	Male	B+	Regular	
20	PAT-1019	Mary	Johnson	1988-12-2	37	Adult	Male	A+	Regular	
21	PAT-1020	Patricia	Garcia	1957-05-0	69	Senior Citizer	Male	AB-	Checkup Needed	
22	PAT-1021	Priya	Jones	1994-06-2	32	Adult	Female	A+	Regular	
23	PAT-1022	Patricia	Singh	1967-11-1	58	Adult	Male	O+	Regular	
24	PAT-1023	Priya	Brown	1962-05-0	64	Senior Citizer	Female	B+	Checkup Needed	
25	PAT-1024	John	Garcia	1992-12-2	33	Adult	Male	A-	Regular	

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1	ContactID	PatientID	Patient Name	Address	Phone	Email	Door No	Address
2	CON-1	PAT-1001	Jennifer	9728 Main St	555-782-3303	patient.1001@example.com	9728	Different
3	CON-2	PAT-1002	Mary	9563 Main St	555-511-5916	patient.1002@example.com	9563	Different
4	CON-3	PAT-1003	John	905 Main St	555-509-2064	patient.1003@example.com	905	Different
5	CON-4	PAT-1004	Aarav	2161 Main St	555-275-5127	patient.1004@example.com	2161	Different
6	CON-5	PAT-1005	Priya	1181 Main St	555-822-1549	patient.1005@example.com	1181	Same
7	CON-6	PAT-1006	James	8087 Main St	555-501-6502	patient.1006@example.com	8087	Same
8	CON-7	PAT-1007	Jennifer	2970 Main St	555-595-3773	patient.1007@example.com	2970	Different
9	CON-8	PAT-1008	Mary	2201 Main St	555-528-9429	patient.1008@example.com	2201	Different
10	CON-9	PAT-1009	John	4212 Main St	555-656-7339	patient.1009@example.com	4212	Different
11	CON-10	PAT-1010	Aarav	2829 Main St	555-185-8212	patient.1010@example.com	2829	Different
12	CON-11	PAT-1011	Mary	4279 Main St	555-964-7026	patient.1011@example.com	4279	Different
13	CON-12	PAT-1012	James	7059 Main St	555-321-6399	patient.1012@example.com	7059	Different
14	CON-13	PAT-1013	John	9185 Main St	555-189-8378	patient.1013@example.com	9185	Different
15	CON-14	PAT-1014	Patricia	3503 Main St	555-410-7909	patient.1014@example.com	3503	Different
16	CON-15	PAT-1015	Michael	6272 Main St	555-352-4555	patient.1015@example.com	6272	Different
17	CON-16	PAT-1016	James	2795 Main St	555-993-2986	patient.1016@example.com	2795	Same
18	CON-17	PAT-1017	James	5242 Main St	555-998-4017	patient.1017@example.com	5242	Different
19	CON-18	PAT-1018	Robert	8905 Main St	555-422-1838	patient.1018@example.com	8905	Different
20	CON-19	PAT-1019	Mary	8677 Main St	555-543-9113	patient.1019@example.com	8677	Same
21	CON-20	PAT-1020	Patricia	5363 Main St	555-883-1913	patient.1020@example.com	5363	Different
22	CON-21	PAT-1021	Priya	8798 Main St	555-151-6507	patient.1021@example.com	8798	Different
23	CON-22	PAT-1022	Patricia	517 Main St	555-876-8411	patient.1022@example.com	517	Different
24	CON-23	PAT-1023	Priya	3216 Main St	555-478-9450	patient.1023@example.com	3216	Different
25	CON-24	PAT-1024	John	7456 Main St	555-253-9254	patient.1024@example.com	7456	Different

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1											
2											
3		1. Basic Contact Stats			3. Email Analysis						
4											
5		Total Contacts		1000	Extract Domain		gives example.com for all				
6		Patients with Contact I		1000	Validate Email Format		Valid				
7		(should match Patients_Master count i				Extract Patient Number from Em		gives 1001, 1002, etc			
8											
9		2. Address Analysis			4. Link to Patients_Master						
10		Extract House Number from Address		Get Patient Name + Contact together							
11		gives 9728, 9563, etc		9728	→ First Name		→ BloodType				
12		Count Patients on "Main St"									
13		" _Main St _"		1000	5. Data Quality Checks						
14		Find Duplicate Addresses			Missing Data Check		Complete				
15		Use Conditional Format Highlight Duplic				Duplicate PatientID Check		IF(COUNTIF(B:B, B2)>1, "Duplicate PatientID", "")			
16		formula: =IF(COUNTIF(C:C, C2)>1, "Same One", "Different")									
17											
18											

	A	B	C	D	E	F	G	H	I	J	K	L
1	PatientID	Encoun	EncounterDate	ProviderID	Provider Name	FacilityID	EncounterType	ICD Code	CPT Code	ChargeAmount	ICD - CPT	CPT Cost Status
2	PAT-1001	ENC-1	2022-03-03	PROV-112	Dr. James Smith	CLI-01	Hospital Admission	I10	85025	264.07	85025 - 264.07	Normal
3	PAT-1001	ENC-2	2022-06-27	PROV-104	Dr. Aarav Smith	CLI-01	Hospital Admission	M54.5	85025	375.43	85025 - 375.43	Normal
4	PAT-1001	ENC-3	2025-02-17	PROV-118	Dr. Robert Brown	CLI-01	Office Visit	J06.9	99213	103.82	99213 - 103.82	Normal
5	PAT-1001	ENC-4	2023-07-27	PROV-105	Dr. Michael Singh	HOS-01	Hospital Admission	J06.9	80053	122.64	80053 - 122.64	Normal
6	PAT-1001	ENC-5	2024-12-19	PROV-101	Dr. Mary Garcia	CLI-01	Telehealth	M54.5	99203	368.7	99203 - 368.7	Normal
7	PAT-1001	ENC-6	2023-05-09	PROV-122	Dr. Linda Singh	CLI-01	Office Visit	E11.9	99203	440.92	99203 - 440.92	High Cost
8	PAT-1001	ENC-7	2025-04-14	PROV-106	Dr. James Brown	HOS-01	Office Visit	J06.9	99213	301.58	99213 - 301.58	Normal
9	PAT-1001	ENC-8	2023-06-20	PROV-116	Dr. Linda Johnson	CLI-01	Emergency	M54.5	99213	365.66	99213 - 365.66	Normal
10	PAT-1001	ENC-9	2022-10-26	PROV-111	Dr. Patricia Smith	HOS-01	Emergency	E11.9	99213	456.05	99213 - 456.05	High Cost
11	PAT-1002	ENC-10	2025-02-22	PROV-104	Dr. Aarav Smith	CLI-01	Office Visit	I10	80053	297.6	80053 - 297.6	Normal
12	PAT-1002	ENC-11	2022-02-03	PROV-106	Dr. James Brown	CLI-01	Hospital Admission	K21.9	85025	124.26	85025 - 124.26	Normal
13	PAT-1002	ENC-12	2024-06-10	PROV-114	Dr. Robert Singh	CLI-01	Telehealth	I10	80053	316.2	80053 - 316.2	Normal
14	PAT-1002	ENC-13	2024-02-17	PROV-130	Dr. Linda Singh	CLI-01	Office Visit	M54.5	85025	228.51	85025 - 228.51	Normal
15	PAT-1002	ENC-14	2022-04-23	PROV-130	Dr. Linda Singh	CLI-01	Office Visit	K21.9	99213	53.34	99213 - 53.34	Normal
16	PAT-1002	ENC-15	2025-02-29	PROV-109	Dr. Patricia Garcia	CLI-01	Hospital Admission	J06.9	99213	413.76	99213 - 413.76	High Cost
17	PAT-1002	ENC-16	2026-04-07	PROV-122	Dr. Linda Singh	CLI-01	Emergency	E11.9	99203	467.05	99203 - 467.05	High Cost
18	PAT-1003	ENC-17	2025-06-21	PROV-107	Dr. John Jones	CLI-01	Hospital Admission	J06.9	85025	116.75	85025 - 116.75	Normal
19	PAT-1003	ENC-18	2025-01-24	PROV-103	Dr. John Singh	HOS-01	Hospital Admission	J06.9	99213	405.15	99213 - 405.15	High Cost
20	PAT-1003	ENC-19	2024-05-21	PROV-110	Dr. James Singh	CLI-01	Telehealth	J06.9	80053	393.69	80053 - 393.69	Normal
21	PAT-1003	ENC-20	2024-03-14	PROV-124	Dr. Priya Garcia	CLI-01	Hospital Admission	J06.9	80053	429.14	80053 - 429.14	High Cost
22	PAT-1004	ENC-21	2022-08-22	PROV-111	Dr. Patricia Smith	CLI-01	Hospital Admission	K21.9	99203	441.43	99203 - 441.43	High Cost
23	PAT-1004	ENC-22	2024-02-16	PROV-104	Dr. Aarav Smith	HOS-01	Office Visit	E11.9	85025	166.47	85025 - 166.47	Normal
24	PAT-1004	ENC-23	2022-09-22	PROV-118	Dr. Robert Brown	CLI-01	Office Visit	I10	80053	489.38	80053 - 489.38	High Cost

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3	EncounterType	1. Revenue & Volume Analysis			3. Clinical Coding Analysis				5. Useful Functions to Add Insights			
4	Emergency	Total Charges by Provider			Most Common ICD Codes				UNIQUE:			
5	Hospital Admission	"Dr. James Smith"	99813.84		ICD Codes "J06.9"				1076	Get list of all providers without du	30	
6	Office Visit	Average Charge per Encounter Type	270.3875437		Most Common CPT Codes				MAX/MIN			
7	Telehealth	"Hospital Admission			(Same with column H, 99203 and 99213 look frequent here = Office V				Highest charge		499.95	
8		Count of Encounters			Average Charge by ICD Code "J06.9"				270.8147305	lowest charge	50.05	
9		"Dr. Linda Singh"	383						XLOOKUP			
10		"Emergency"	1335						Lookup provider charges by name		405.15	
11									TEXTJOIN			
12	FacilityID	2. Provider & Facility Performance			4. Encounter Type Breakdown				Combine ICD + CPT for a summary		I10 - 85025	
13	CLI-01	Revenue by Facility			% of Each Encounter Type				IF			
14	HOS-01	Compare Clinic vs	778815.3		"Office Visit"				26%	Flag high cost cases	=IF(I2>400, "High Cost", "Normal")	
15		Hospital	728833.72		Encounter Type Total				1437			
16		Encounters per Facility							5492			
17		CLI-01(Clinic)	2825									
18		HOS-01(Hospital)	2666									
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1	PrescriptionID	EncounterID	DatePrescribed	CPT Code	Charges	MedicationID	Medication Name	PharmacyID	Pharmacy Name	Year	Month	MMM-YYYY	RX Cost Statu
2	RX-1	ENC-7	2024-06-02	99213	301.58	MED03	Amoxicillin 250mg	PHARM-01	CVS Pharmacy	2024	6	Jun-2024	Normal
3	RX-2	ENC-8	2022-05-02	99213	365.66	MED01	Lisinopril 10mg	PHARM-02	Walgreens	2022	5	May-2022	Normal
4	RX-3	ENC-9	2025-10-08	99213	456.05	MED03	Amoxicillin 250mg	PHARM-04	PNRao Hospital Pharmacy	2025	10	Oct-2025	High Cost RX
5	RX-4	ENC-11	2022-09-12	85025	124.26	MED03	Amoxicillin 250mg	PHARM-02	Walgreens	2022	9	Sep-2022	Normal
6	RX-5	ENC-12	2026-05-16	80053	316.2	MED04	Ibuprofen 200mg	PHARM-03	Rite Aid	2026	5	May-2026	Normal
7	RX-6	ENC-13	2025-01-18	85025	228.51	MED04	Ibuprofen 200mg	PHARM-03	Rite Aid	2025	1	Jan-2025	Normal
8	RX-7	ENC-14	2022-03-17	99213	53.34	MED03	Amoxicillin 250mg	PHARM-04	PNRao Hospital Pharmacy	2022	3	Mar-2022	Normal
9	RX-8	ENC-18	2022-07-18	99213	405.15	MED02	Metformin 500mg	PHARM-01	CVS Pharmacy	2022	7	Jul-2022	High Cost RX
10	RX-9	ENC-19	2024-01-19	80053	393.69	MED02	Metformin 500mg	PHARM-04	PNRao Hospital Pharmacy	2024	1	Jan-2024	Normal
11	RX-10	ENC-20	2024-05-15	80053	429.14	MED02	Metformin 500mg	PHARM-02	Walgreens	2024	5	May-2024	High Cost RX
12	RX-11	ENC-22	2024-08-03	85025	166.47	MED01	Lisinopril 10mg	PHARM-03	Rite Aid	2024	8	Aug-2024	Normal
13	RX-12	ENC-24	2022-01-28	99213	113.6	MED04	Ibuprofen 200mg	PHARM-04	PNRao Hospital Pharmacy	2022	1	Jan-2022	Normal
14	RX-13	ENC-25	2022-04-03	85025	498.28	MED04	Ibuprofen 200mg	PHARM-04	PNRao Hospital Pharmacy	2022	4	Apr-2022	High Cost RX
15	RX-14	ENC-26	2025-08-11	85025	92.36	MED03	Amoxicillin 250mg	PHARM-01	CVS Pharmacy	2025	8	Aug-2025	Normal
16	RX-15	ENC-30	2023-04-21	99203	70.8	MED03	Amoxicillin 250mg	PHARM-02	Walgreens	2023	4	Apr-2023	Normal
17	RX-16	ENC-31	2025-03-12	85025	491.99	MED03	Amoxicillin 250mg	PHARM-01	CVS Pharmacy	2025	3	Mar-2025	High Cost RX
18	RX-17	ENC-39	2022-04-20	99213	465.82	MED01	Lisinopril 10mg	PHARM-02	Walgreens	2022	4	Apr-2022	High Cost RX
19	RX-18	ENC-40	2022-05-11	80053	378.64	MED04	Ibuprofen 200mg	PHARM-01	CVS Pharmacy	2022	5	May-2022	Normal
20	RX-19	ENC-41	2026-06-25	85025	206.96	MED03	Amoxicillin 250mg	PHARM-04	PNRao Hospital Pharmacy	2026	6	Jun-2026	Normal
21	RX-20	ENC-42	2024-07-04	99213	76.28	MED01	Lisinopril 10mg	PHARM-03	Rite Aid	2024	7	Jul-2024	Normal
22	RX-21	ENC-44	2023-05-06	99203	96.6	MED04	Ibuprofen 200mg	PHARM-01	CVS Pharmacy	2023	5	May-2023	Normal
23	RX-22	ENC-47	2026-01-20	99213	449.09	MED02	Metformin 500mg	PHARM-02	Walgreens	2026	1	Jan-2026	High Cost RX
24	RX-23	ENC-48	2025-05-09	85025	211.08	MED04	Ibuprofen 200mg	PHARM-03	Rite Aid	2025	5	May-2025	Normal
25	RX-24	ENC-50	2022-07-22	99213	398.95	MED04	Ibuprofen 200mg	PHARM-04	PNRao Hospital Pharmacy	2022	7	Jul-2022	Normal

	A	B	C	D	E	F	G	H	I	J	K
1	DiagnosisID	EncounterID	ICD10_Code	DiagnosisDescription	Patient_ID	Age	Column1	CPT_Code	Medication_Name	Category	Condition
2	DIAG-1	ENC-1	I10	Essential (primary) hypertension	PAT-1001	52 Adult			0	0 I	Chronic
3	DIAG-2	ENC-2	M54.5	Low back pain	PAT-1001	52 Adult			0	0 M	Acute
4	DIAG-3	ENC-3	J06.9	Acute upper respiratory infection, unspecified	PAT-1001	52 Adult			0	0 J	Acute
5	DIAG-4	ENC-4	J06.9	Acute upper respiratory infection, unspecified	PAT-1001	52 Adult			0	0 J	Acute
6	DIAG-5	ENC-5	M54.5	Low back pain	PAT-1001	52 Adult			0	0 M	Acute
7	DIAG-6	ENC-6	E11.9	Type 2 diabetes mellitus without complications	PAT-1001	52 Adult			0	0 E	Chronic
8	DIAG-7	ENC-7	J06.9	Acute upper respiratory infection, unspecified	PAT-1001	52 Adult		99213 Amoxicillin 250mg			Acute
9	DIAG-8	ENC-8	M54.5	Low back pain	PAT-1001	52 Adult		99213 Lisinopril 10mg		M	Acute
10	DIAG-9	ENC-9	E11.9	Type 2 diabetes mellitus without complications	PAT-1001	52 Adult		99213 Amoxicillin 250mg		E	Chronic
11	DIAG-10	ENC-10	I10	Essential (primary) hypertension	PAT-1002	30 young		0		0 I	Chronic
12	DIAG-11	ENC-11	K21.9	Gastro-esophageal reflux disease without esophagitis	PAT-1002	30 young		85025 Amoxicillin 250mg		K	Acute
13	DIAG-12	ENC-12	I10	Essential (primary) hypertension	PAT-1002	30 young		80053 Ibuprofen 200mg		I	Chronic
14	DIAG-13	ENC-13	M54.5	Low back pain	PAT-1002	30 young		85025 Ibuprofen 200mg		M	Acute
15	DIAG-14	ENC-14	K21.9	Gastro-esophageal reflux disease without esophagitis	PAT-1002	30 young		99213 Amoxicillin 250mg		K	Acute
16	DIAG-15	ENC-15	J06.9	Acute upper respiratory infection, unspecified	PAT-1002	30 young		0		0 J	Acute
17	DIAG-16	ENC-16	E11.9	Type 2 diabetes mellitus without complications	PAT-1002	30 young		0		0 E	Chronic
18	DIAG-17	ENC-17	J06.9	Acute upper respiratory infection, unspecified	PAT-1003	59 Adult		0		0 J	Acute
19	DIAG-18	ENC-18	J06.9	Acute upper respiratory infection, unspecified	PAT-1003	59 Adult		99213 Metformin 500mg		J	Acute
20	DIAG-19	ENC-19	J06.9	Acute upper respiratory infection, unspecified	PAT-1003	59 Adult		80053 Metformin 500mg		J	Acute
21	DIAG-20	ENC-20	J06.9	Acute upper respiratory infection, unspecified	PAT-1003	59 Adult		80053 Metformin 500mg		J	Acute
22	DIAG-21	ENC-21	K21.9	Gastro-esophageal reflux disease without esophagitis	PAT-1004	55 Adult		0		0 K	Acute
23	DIAG-22	ENC-22	E11.9	Type 2 diabetes mellitus without complications	PAT-1004	55 Adult		85025 Lisinopril 10mg		E	Chronic
24	DIAG-23	ENC-23	I10	Essential (primary) hypertension	PAT-1004	55 Adult		0		0 I	Chronic
25	DIAG-24	ENC-24	I10	Essential (primary) hypertension	PAT-1004	55 Adult		99213 Ibuprofen 200mg		I	Chronic
26	DIAG-25	ENC-25	J06.9	Acute upper respiratory infection, unspecified	PAT-1004	55 Adult		85025 Ibuprofen 200mg		J	Acute
27	DIAG-26	ENC-26	M54.5	Low back pain	PAT-1005	18 young		85025 Amoxicillin 250mg		M	Acute
28	DIAG-27	ENC-27	E11.9	Type 2 diabetes mellitus without complications	PAT-1005	18 young		0		0 E	Chronic

	M	N	O	P	Q	R	S	T	U	V	W	X
1												
2												
3												
4												
5												
6												
7												
8												
9												
10												
11												
12												
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16												
17												
18												
19												
20												
21												
22												
23												
24												

	A	B	C	D	E	F	G	H	I
1	ClaimID	ChargeID	InsuranceProvider	AmountBilled	AmountPaid	Remaining	Status	Balance	Categorize claim size
2	CLM-1	CHG-1	Aetna	422.53	376.0517	46.4783	Paid	OK	High Value
3	CLM-2	CHG-2	Aetna	331.74	265.392	66.348	Paid	OK	Standard
4	CLM-3	CHG-3	Aetna	190.56	165.7872	24.7728	Paid	OK	Standard
5	CLM-4	CHG-4	Aetna	183.79	161.7352	22.0548	Paid	OK	Standard
6	CLM-5	CHG-5	Aetna	263.66	234.6574	29.0026	Paid	OK	Standard
7	CLM-6	CHG-6	Aetna	427.37	358.9908	68.3792	Paid	OK	High Value
8	CLM-7	CHG-7	Aetna	396.93	321.5133	75.4167	Paid	OK	Standard
9	CLM-8	CHG-8	Aetna	346.53	291.0852	55.4448	Paid	OK	Standard
10	CLM-9	CHG-9	Aetna	463.75	338.5375	125.2125	Paid	Follow Up	High Value
11	CLM-10	CHG-10	Aetna	71.28	60.588	10.692	Paid	OK	Standard
12	CLM-11	CHG-11	Aetna	158.57	130.0274	28.5426	Paid	OK	Standard
13	CLM-12	CHG-12	Aetna	498.93	444.0477	54.8823	Paid	OK	High Value
14	CLM-13	CHG-13	Aetna	121.36	104.3696	16.9904	Paid	OK	Standard
15	CLM-14	CHG-14	Aetna	302.8	242.24	60.56	Paid	OK	Standard
16	CLM-15	CHG-15	Aetna	394.06	279.7826	114.2774	Paid	Follow Up	Standard
17	CLM-16	CHG-17	Aetna	347.49	302.3163	45.1737	Paid	OK	Standard
18	CLM-17	CHG-18	Aetna	468.47	332.6137	135.8563	Paid	Follow Up	High Value
19	CLM-18	CHG-19	Aetna	379.02	265.314	113.706	Paid	Follow Up	Standard
20	CLM-19	CHG-21	Aetna	138.89	108.3342	30.5558	Paid	OK	Standard
21	CLM-20	CHG-23	Aetna	363.12	279.6024	83.5176	Paid	OK	Standard
22	CLM-21	CHG-24	Aetna	54.73	43.2367	11.4933	Paid	OK	Standard
23	CLM-22	CHG-25	Aetna	111.56	85.9012	25.6588	Paid	OK	Standard
24	CLM-23	CHG-26	Aetna	315.22	283.698	31.522	Paid	OK	Standard
25	CLM-24	CHG-27	Aetna	497.95	403.3395	94.6105	Paid	OK	High Value

	K	L	M	N	O	P	Q
1							
2	1. Revenue & Payment Analysis				4. Data Quality & Validation Checks		
3							
4	Total Amount Billed				Flag High Balance Claims	OK	
5	Total revenue submitted to insurance		1386951				
6	Total Amount Paid				5. Functions for this sheet		
7	What insurance actually paid		1109467				
8	Total Outstanding/Remaining				LARGE		499.95
9	Patient/Insurance balance left		277483.8		SMALL		50.07
10	Average Payment per Claim						
11	AVERAGE		218.7004				
12	Collection Rate %						
13	How much of billed is collected		80%				
14							
15	2. Claim Status Analysis						
16							
17	Count of Paid Claims						
18	Paid		5073				
19	Highest Outstanding Balance						
20	MAX		148.932				
21							
22	3. Insurance Provider Analysis						
23							
24	Total Billed by Provider		1386951				
25	Average Paid by Provider		218.7004				
26	Payment Rate by Provider		0.799933				
27							

	A	B	C	D	E	F	G	H	I
1	ChargeID	EncounterID	CPT_Code	ChargeAmount	Charge	Diagnosis_Description	Procedure_Description	Provide	Categorize service type
2	CHG-1	ENC-1	85025	264.07	Normal	Essential (primary) hypertension	Complete blood count (CBC)	PROV-112 Lab	
3	CHG-2	ENC-2	85025	375.43	High	Low back pain	Comprehensive metabolic panel	PROV-104 Lab	
4	CHG-3	ENC-3	99213	103.82	Normal	Acute upper respiratory infection, unspecified	Office visit	PROV-118 E&M Visit	
5	CHG-4	ENC-4	80053	122.64	Normal	Acute upper respiratory infection, unspecified	Office visit	PROV-105 Lab	
6	CHG-5	ENC-5	99203	368.7	High	Low back pain	Complete blood count (CBC)	PROV-101 E&M Visit	
7	CHG-6	ENC-6	99203	440.92	High	Type 2 diabetes mellitus without complications	Office visit	PROV-122 E&M Visit	
8	CHG-7	ENC-7	99213	301.58	Normal	Acute upper respiratory infection, unspecified	Complete blood count (CBC)	PROV-106 E&M Visit	
9	CHG-8	ENC-8	99213	365.66	High	Low back pain	Complete blood count (CBC)	PROV-116 E&M Visit	
10	CHG-9	ENC-9	99213	456.05	High	Type 2 diabetes mellitus without complications	Complete blood count (CBC)	PROV-111 E&M Visit	
11	CHG-10	ENC-10	80053	297.6	Normal	Essential (primary) hypertension	Comprehensive metabolic panel	PROV-104 Lab	
12	CHG-11	ENC-11	85025	124.26	Normal	Gastro-esophageal reflux disease without esophagitis	Office visit	PROV-106 Lab	
13	CHG-12	ENC-12	80053	316.2	Normal	Essential (primary) hypertension	Comprehensive metabolic panel	PROV-114 Lab	
14	CHG-13	ENC-13	85025	228.51	Normal	Low back pain	Complete blood count (CBC)	PROV-130 Lab	
15	CHG-14	ENC-14	99213	53.34	Normal	Gastro-esophageal reflux disease without esophagitis	Office visit	PROV-130 E&M Visit	
16	CHG-15	ENC-15	99213	413.76	High	Acute upper respiratory infection, unspecified	Office visit	PROV-109 E&M Visit	
17	CHG-16	ENC-16	99203	467.05	High	Type 2 diabetes mellitus without complications	Office visit	PROV-122 E&M Visit	
18	CHG-17	ENC-17	85025	116.75	Normal	Acute upper respiratory infection, unspecified	Complete blood count (CBC)	PROV-107 Lab	
19	CHG-18	ENC-18	99213	405.15	High	Acute upper respiratory infection, unspecified	Comprehensive metabolic panel	PROV-103 E&M Visit	
20	CHG-19	ENC-19	80053	393.69	High	Acute upper respiratory infection, unspecified	Comprehensive metabolic panel	PROV-110 Lab	
21	CHG-20	ENC-20	80053	429.14	High	Acute upper respiratory infection, unspecified	Office visit	PROV-124 Lab	
22	CHG-21	ENC-21	99203	441.43	High	Gastro-esophageal reflux disease without esophagitis	Comprehensive metabolic panel	PROV-111 E&M Visit	
23	CHG-22	ENC-22	85025	166.47	Normal	Type 2 diabetes mellitus without complications	Office visit	PROV-104 Lab	
24	CHG-23	ENC-23	80053	489.38	High	Essential (primary) hypertension	Comprehensive metabolic panel	PROV-118 Lab	
25	CHG-24	ENC-24	99213	113.6	Normal	Essential (primary) hypertension	Office visit	PROV-111 E&M Visit	

85025 = CBC					3. Pricing Analysis				
80053 = CMP					Highest Lowest Charges				
99213 = Office Visit Established					Highest Charge	499.95			
99203 = Office Visit New					Lowest Charge	50.05			
					Price Variance by CPT				
1. Revenue Analysis by CPT					Range for CBC	449.42			
					Flag Outliers	Normal			
Total Revenue									
All charges combined			1541475		4. Cross-Analysis with Other Sheets				
Total by CPT Code									
CBC Revenue			370715.3		Revenue by Diagnosis		DiagnosisDescription		
CMP Revenue			403607.4		Revenue by Procedure		ProcedureDescription		
Est. Visit Revenue			384639.8		Who is billing the most		ProviderID		
New Visit Revenue			382512.5						
Average Charge per CPT					5. Functions for this sheet				
Avg for New Patient Visit			277.384						
					UNIQUE	85025	List of all CPT codes		
2. Volume / Utilization Analysis						99213			
						80053			
Count of Each CPT						99203			
	85025		1369		IF		Categorize service type		
	99213		1401						
% Mix of Services									
% of CBCs			24%						

CPT_Code

80053

85025

99203

99213

	A	B	C	D	E	F
1	ProviderID	FullName	Specialty	Provider	Column1	Column2
2	PROV-101	Dr. Mary Garcia	General Practice	Garcia	OK	Dr. Mary Garcia - General Practice
3	PROV-102	Dr. Michael Brown	Oncology	Brown	High Staffir	Dr. Michael Brown - Oncology
4	PROV-103	Dr. John Singh	Dermatology	Singh	OK	Dr. John Singh - Dermatology
5	PROV-104	Dr. Aarav Smith	Dermatology	Smith	OK	Dr. Aarav Smith - Dermatology
6	PROV-105	Dr. Michael Singh	Oncology	Singh	High Staffir	Dr. Michael Singh - Oncology
7	PROV-106	Dr. James Brown	General Practice	Brown	OK	Dr. James Brown - General Practice
8	PROV-107	Dr. John Jones	Oncology	Jones	High Staffir	Dr. John Jones - Oncology
9	PROV-108	Dr. Michael Williams	Cardiology	Williams	OK	Dr. Michael Williams - Cardiology
10	PROV-109	Dr. Patricia Garcia	Gastroenterology	Garcia	OK	Dr. Patricia Garcia - Gastroenterology
11	PROV-110	Dr. James Singh	Neurology	Singh	OK	Dr. James Singh - Neurology
12	PROV-111	Dr. Patricia Smith	Pediatrics	Smith	OK	Dr. Patricia Smith - Pediatrics
13	PROV-112	Dr. James Smith	Dermatology	Smith	OK	Dr. James Smith - Dermatology
14	PROV-113	Dr. James Jones	Pediatrics	Jones	OK	Dr. James Jones - Pediatrics
15	PROV-114	Dr. Robert Singh	Neurology	Singh	OK	Dr. Robert Singh - Neurology
16	PROV-115	Dr. Mary Williams	Cardiology	Williams	OK	Dr. Mary Williams - Cardiology
17	PROV-116	Dr. Linda Johnson	Oncology	Johnson	High Staffir	Dr. Linda Johnson - Oncology
18	PROV-117	Dr. John Singh	Pediatrics	Singh	OK	Dr. John Singh - Pediatrics
19	PROV-118	Dr. Robert Brown	Oncology	Brown	High Staffir	Dr. Robert Brown - Oncology
20	PROV-119	Dr. James Smith	Cardiology	Smith	OK	Dr. James Smith - Cardiology
21	PROV-120	Dr. Mary Johnson	Gastroenterology	Johnson	OK	Dr. Mary Johnson - Gastroenterology
22	PROV-121	Dr. Jennifer Garcia	Cardiology	Garcia	OK	Dr. Jennifer Garcia - Cardiology
23	PROV-122	Dr. Linda Singh	Neurology	Singh	OK	Dr. Linda Singh - Neurology
24	PROV-123	Dr. James Singh	Oncology	Singh	High Staffir	Dr. James Singh - Oncology
25	PROV-124	Dr. Priya Garcia	Gastroenterology	Garcia	OK	Dr. Priya Garcia - Gastroenterology

	G	H	I	J	K	L	M	N	O	P	Q
2	Specialty	1. Workforce / Specialty Analysis			3. Staffing Gaps Analysis						
3											
4	Cardiology	Total Providers			Flag Overrepresented Specialty			OK			
5	Dermatology	Providers		30							
6	Gastroenterology	Count by Specialty			4. Functions for this sheet						
7	General Practice	Oncology		6							
8	Neurology	Cardiology		5	LEFT/RIGHT:			Dr.			
9	Oncology	Dermatology		3	TEXTJOIN			Dr. Mary Garcia - General Practice			
10	Pediatrics	Pediatrics		5	SORT/UNIQUE			=SORT(UNIQUE(C2:C25))			
11		General Practice		3				Cardiology			
12		Neurology		4				Dermatology			
13		Gastroenterology		4				Gastroenterology			
14		% Distribution by Specialty						General Practice			
15		Oncology		19%				Neurology			
16								Oncology			
17		2. Name / Demographics Analysis						Pediatrics			
18											
19		Count by Last Name		Extracts Last	Oncology: 5 providers - Largest department						
20		how many Singhs		8	Cardiology: 4 providers						
21				4	Dermatology, Pediatrics, Neurology, Gastroenterology: 3 each						
22		Count of Dr. Michael		3	General Practice: 2 providers - Smallest						
23		List Unique Specialties		General Practice							
24				Oncology							
25				Dermatology							
26				Cardiology							
27				Gastroenterology							
28				Neurology							
29				Pediatrics							

	A	B	C	D	E	F	G	H	I	J	K	L
1	VitalsID	EncounterID	SystolicBP	DiastolicBP	HeartRate	Temperature_F	Blood	Risk St	Categ	Patient ID	Age_group	DiagnosisDescription
2	VIT-1	ENC-1	134	79	74	97.7	Normal	Stable	Stage 1 HT	PAT-1001	Adult	Essential (primary) hypertension
3	VIT-2	ENC-2	110	73	68	98.6	Normal	Stable	Normal	PAT-1001	Adult	Low back pain
4	VIT-3	ENC-3	137	83	89	98.3	Normal	Stable	Stage 1 HT	PAT-1001	Adult	Acute upper respiratory infection, unspecified
5	VIT-4	ENC-4	127	75	76	98.1	Normal	Stable	Elevated	PAT-1001	Adult	Acute upper respiratory infection, unspecified
6	VIT-5	ENC-5	139	72	98	98.4	Normal	High Risk	Stage 1 HT	PAT-1001	Adult	Low back pain
7	VIT-6	ENC-6	137	84	82	98.4	Normal	Stable	Stage 1 HT	PAT-1001	Adult	Type 2 diabetes mellitus without complications
8	VIT-7	ENC-7	128	73	63	99.4	Normal	Stable	Elevated	PAT-1001	Adult	Acute upper respiratory infection, unspecified
9	VIT-8	ENC-8	112	75	68	98.7	Normal	Stable	Normal	PAT-1001	Adult	Low back pain
10	VIT-9	ENC-9	135	72	78	98.3	Normal	Stable	Stage 1 HT	PAT-1001	Adult	Type 2 diabetes mellitus without complications
11	VIT-10	ENC-10	110	75	63	99.4	Normal	Stable	Normal	PAT-1002	young	Essential (primary) hypertension
12	VIT-11	ENC-11	140	80	65	98.5	Hypertens	Stable	Stage 2 HT	PAT-1002	young	Gastro-esophageal reflux disease without esophagitis
13	VIT-12	ENC-12	129	86	100	99.1	Normal	Stable	Elevated	PAT-1002	young	Essential (primary) hypertension
14	VIT-13	ENC-13	113	82	81	99.4	Normal	Stable	Normal	PAT-1002	young	Low back pain
15	VIT-14	ENC-14	113	70	90	98.6	Normal	Stable	Normal	PAT-1002	young	Gastro-esophageal reflux disease without esophagitis
16	VIT-15	ENC-15	129	83	77	98	Normal	Stable	Elevated	PAT-1002	young	Acute upper respiratory infection, unspecified
17	VIT-16	ENC-16	111	77	66	98.5	Normal	Stable	Normal	PAT-1002	young	Type 2 diabetes mellitus without complications
18	VIT-17	ENC-17	112	82	82	97.6	Normal	Stable	Normal	PAT-1003	Adult	Acute upper respiratory infection, unspecified
19	VIT-18	ENC-18	123	75	66	97.6	Normal	Stable	Elevated	PAT-1003	Adult	Acute upper respiratory infection, unspecified
20	VIT-19	ENC-19	135	79	99	97.7	Normal	High Risk	Stage 1 HT	PAT-1003	Adult	Acute upper respiratory infection, unspecified
21	VIT-20	ENC-20	134	77	60	97.9	Normal	Stable	Stage 1 HT	PAT-1003	Adult	Acute upper respiratory infection, unspecified
22	VIT-21	ENC-21	140	77	96	98.2	Hypertens	High Risk	Stage 2 HT	PAT-1004	Adult	Gastro-esophageal reflux disease without esophagitis
23	VIT-22	ENC-22	139	82	87	98	Normal	Stable	Stage 1 HT	PAT-1004	Adult	Type 2 diabetes mellitus without complications
24	VIT-23	ENC-23	124	90	71	97.6	Hypertens	Stable	Elevated	PAT-1004	Adult	Essential (primary) hypertension
25	VIT-24	ENC-24	138	80	89	97.8	Normal	Stable	Stage 1 HT	PAT-1004	Adult	Essential (primary) hypertension

	N	O	P	Q	R	S	T	U	V
1									
2	1. Basic Vitals Statistics				3. BP Category Analysis				
3									
4	Average Vitals				BP categories per AHA guidelines:				
5	Avg Systolic BP			125.0634	Stage 1 HTN				
6	Avg Diastolic BP			79.85109	Count of Stage 2 HTN			169	
7	Avg Heart Rate			79.96081					
8	Avg Temperature			98.50319	4. Cross-Analysis with Other Sheets				
9	Min/Max Values								
10	BP range			MAX 140	Vitals by Age Group			124.8814	Senior Citizen
11				MIN 110	Vitals by Diagnosis			125.5368	
12	HR range			MAX 100					
13				MIN 60					
14	Standard Deviation								
15	BP variability			8.958375					
16									
17	2. Clinical Flag / Abnormal Vitals Analysis								
18									
19	High Blood Pressure								
20	Count hypertensive patients			423					
21	Fever			Normal					
22	Tachycardia / Bradycardia			Normal					
23	Combined Risk			Stable					
24									

Overall Dashboard:



- %RegularCheckup=DIVIDE(CALCULATE([TotalPatients],Patients_Master[Checkup Routine]="Regular"), [Total Patients])
- ABO Group = Left(Patients_Master[BloodType],LEN(Patients_Master[BloodType])-1)

```

1 Age Band =
2 SWITCH(TRUE(),
3     Patients_Master[Age ] < 20, "0-19",
4     Patients_Master[Age ] < 30, "20-29",
5     Patients_Master[Age ] < 40, "30-39",
6     Patients_Master[Age ] < 50, "40-49",
7     Patients_Master[Age ] < 60, "50-59",
8     "60+"
9 )

```

- Avg Age = AVERAGE(Patients_Master[Age])
- Birth Quarter = "Q" & FORMAT(Patients_Master[DateOfBirth], "Q")
- Birth_Month = Format(Patients_Master[DateOfBirth],"MMMM")
- Birthyear = Year(Patients_Master[DateOfBirth])
- Count O+ = CALCULATE([Total Patients], Patients_Master[Full Blood Group]="O+ Positive")

Table 02: Patient_Contacts Calculated Columns & DAX Measures

- % Verified Address = DIVIDE(CALCULATE([Total Contacts], Patient_Contacts[Address Verifi]="Same"), [Total Contacts])
- Address Match = IF(Patient_Contacts[Address Verifi] = "Same", "Verified", "Needs Review")
- Area Code = LEFT(Patient_Contacts[Phone], 3)
- Total Contacts = COUNTROWS('Patient_Contacts')
- Total of Patients = DISTINCTCOUNT(Patient_Contacts[PatientID])

Table 03: Encounters_Log Calculated Columns & DAX Measures

- % High Cost = DIVIDE(CALCULATE([Total Encounters], Encounters_Log[CPT Cost Status]="High Cost"), [Total Encounters])
- Avg Charge = DIVIDE([Total Revenue], [Total Encounters])
- Charge Tier = SWITCH(TRUE(), Encounters_Log[ChargeAmount] < 430, "Low", Encounters_Log[ChargeAmount] < 470, "Mid", "High")
- Emergency Encounters = CALCULATE([Total Encounters], Encounters_Log[EncounterType]="Emergency")
- Encounter Month = FORMAT(Encounters_Log[EncounterDate], "MMM YYYY")
- Revenue per Patient = DIVIDE([Total Revenue], [Unique Patients])
- Total Encounters = DISTINCTCOUNT(Encounters_Log[EncounterID])
- Total Revenue = SUM(Encounters_Log[ChargeAmount])
- Uniqueof Patients = DISTINCTCOUNT(Encounters_Log[PatientID])

Table 04: Charges Table Calculated Columns & DAX Measures

- Avg Charge Amount = AVERAGE(Charges[ChargeAmount])
- Charge Bucket =
SWITCH(TRUE(),
Charges[ChargeAmount] < 100, "Low <100",
Charges[ChargeAmount] < 250, "Mid 100-249",
"High 250+"
)
- ChargeCategory =
Switch(True(),Charges[ChargeAmount]<100,"Low",Charges[ChargeAmount]<250,"Medium","High")
- Charge Count = COUNTROWS(Charges)
- Charges by High = CALCULATE([Total Charges], Charges[Charge Category] = "High")
- Charges by Low = CALCULATE([Total Charges], Charges[Charge Category] = "Low")
- Charges by Medium = CALCULATE([Total Charges], Charges[Charge Category] = "Medium")
- Charges by Provider Rank =
RANKX(ALL('Charges'[Provider_ID]), [Total Charges])
- Complexity Flag = IF(Charges[Charge Category] = "High", "Complex Case", "Standard Case")
- CPT Group = LEFT(Charges[CPT_Code],2)&"XX"//Group CPT Code
- HighValueCharge=
IF(Charges[ChargeAmount]>AVERAGE(Charges[ChargeAmount]),"Yes","No")
- High Value Count = CALCULATE(COUNTROWS(Charges), Charges[High Value Charge] = "Yes")
- Normal Charge % = DIVIDE(
CALCULATE([Total Charges], 'Charges'[Charge_Status] = "Normal"),
[Total Charges]
)
- Provideer Short = SUBSTITUTE(Charges[Provider_ID],"PROV-","DR")
- Revenue per Encounter = DIVIDE([Total Charges],[Total Encounter])
- Top Diagnosis Revenue =
MAXX(TOPN(1, SUMMARIZE('Charges', 'Charges'[Diagnosis_Description], "Rev", [Total Charges]), [Rev]), 'Charges'[Diagnosis_Description])
- Total Charges = Sum(Charges[ChargeAmount])
- Total Encounter = DISTINCTCOUNT(Charges[EncounterID])
- Total Prescriptions = DISTINCTCOUNT(Prescriptions[PrescriptionID])
- Unique of Patients = DISTINCTCOUNT(Prescriptions[EncounterID])
- Value Label = IF(Charges[High Value Charge] = "Yes", "High Value", "Regular")

Table 05: Diagnoses Table Calculated Columns & DAX Measures

- Diagnoses per Patient = DIVIDE(Total Diagnoses, [Unique Patients])
- ICD Chapter =
SWITCH(LEFT(Diagnoses[ICD10_Code],1),
"I", "Circulatory",
"J", "Respiratory",
"K", "Digestive",
"M", "Musculoskeletal",
"E", "Endocrine",
"Other"
)

- Is Repeat Patient =
VAR Cnt = CALCULATE(COUNTROWS(Diagnoses), ALLEXCEPT(Diagnoses, Diagnoses[Patient_ID]))
RETURN IF(Cnt > 1, "Repeat", "New")
- Top Diagnosis =
VAR Tbl = TOPN(1, SUMMARIZECOLUMNS(Diagnoses[DiagnosisDescription], "Cnt", [Total Diagnoses]), [Cnt])
RETURN SELECTCOLUMNS(Tbl, "Diag", Diagnoses[DiagnosisDescription])
- Total Diagnoses = COUNTROWS(Diagnoses)

Table 06: Insurance_Claims Table Calculated Columns & DAX Measures

- Avg Billed = DIVIDE([Total Billed], [Total Claims])
- Avg Paid = DIVIDE([Total Paid], [Total Claims])
- Claim Size Bucket =
SWITCH(TRUE(),
Insurance_Claims[AmountBilled] < 100, "Small <100",
Insurance_Claims[AmountBilled] < 300, "Medium 100-299",
"Large 300+"
)
- Collection Rate % = DIVIDE([Total Paid], [Total Billed], 0)
- Denial Risk = IF(Insurance_Claims[Payment %] < 0.8, "High Risk", "OK")
- InsurerSize = Insurance_Claims[InsuranceProvider] & " - " & Insurance_Claims[Claim Size Bucket]
- Paid % Standard = CALCULATE([Collection Rate %], Insurance_Claims[Categorize claim size]="Standard")
- Payment%=DIVIDE(Insurance_Claims[AmountPaid], Insurance_Claims[AmountBilled], 0)
- Total Billed = SUM(Insurance_Claims[AmountBilled])
- Total Claims = DISTINCTCOUNT(Insurance_Claims[ClaimID])
- Total Paid = SUM(Insurance_Claims[AmountPaid])
- Total Remaining = SUM(Insurance_Claims[Remaining])
- Write-off = [Total Billed] - [Total Paid] - [Total Remaining]

Table 07: Prescriptions Table Calculated Columns & DAX Measures

- Avg RX Charge = DIVIDE([Total RX Charges], [Total Prescriptions])
- Charge Tier =
SWITCH(TRUE(),
Prescriptions[Charges] < 100, "Low <100",
Prescriptions[Charges] < 250, "Mid 100-249",
"High 250+"
)
- Charges by CVS = CALCULATE([Total RX Charges], Prescriptions[Pharmacy Name]="CVS Pharmacy")
- Medication Category =
SWITCH(TRUE(),
CONTAINSSTRING(Prescriptions[Medication Name], "Amoxicillin"), "Antibiotic",
CONTAINSSTRING(Prescriptions[Medication Name], "Metformin"), "Diabetes",
CONTAINSSTRING(Prescriptions[Medication Name], "Lisinopril"), "Blood Pressure",
CONTAINSSTRING(Prescriptions[Medication Name], "Ibuprofen"), "Pain Relief",
"Other"
)

- Pharmacy Chain =
SWITCH(Prescriptions[Pharmacy Name],
"CVS Pharmacy", "CVS",
"Walgreens", "Walgreens",
"Rite Aid", "Rite Aid",
"PNRao Hospital Pharmacy", "Hospital",
"Other"
)
- Pharmacy Type =
IF(CONTAINSSTRING(Prescriptions[Pharmacy Name], "Hospital"), "Hospital",
"Retail")
- Prescription Month = FORMAT(Prescriptions[DatePrescribed], "MMM YYYY")
- Top Med Charges =
MAXX(TOPN(1, SUMMARIZECOLUMNS(Prescriptions[Medication Name], "TC",
[Total RX Charges]), [TC]), [TC])
- Total RX Charges = SUM(Prescriptions[Charges])
- Unique Medications = DISTINCTCOUNT(Prescriptions[Medication Name])
- Unique Pharmacies = DISTINCTCOUNT(Prescriptions[Pharmacy Name])

Table 08: Procedures Table Calculated Columns & DAX Measures

- % Lab Work = DIVIDE(CALCULATE([Total Procedures], Procedures[Procedure Categorize]="Lab Work"), [Total Procedures])
- Avg Duration = AVERAGE(Procedures[Minutes])
- Inpatient Count = CALCULATE([Total Procedures], Procedures[Patient Type]="Inpatient")
- Is Lab = IF(Procedures[Procedure Categorize] = "Lab Work", "Yes", "No")
- Procedure Group =
SWITCH(TRUE(),
CONTAINSSTRING(Procedures[Procedure Description], "blood"), "Lab - Blood",
CONTAINSSTRING(Procedures[procedure Description], "x-ray"), "Imaging",
CONTAINSSTRING(Procedures[procedure Description], "surgery"), "Surgery",
Procedures[Procedure Categorize]
)
- Procedures per Encounter = DIVIDE([Total Procedures],
DISTINCTCOUNT(Procedures[EncounterID]))
- Total Minutes = SUM(Procedures[Minutes])
- Total Procedures = COUNTROWS(Procedures)

Table 09 : Providers_Master Table Calculated Columns & DAX Measures

- % High Staffing = DIVIDE([High Staffing Count], [Total Providers])
 - High Staffing Count = CALCULATE(COUNTROWS(Providers_Master),
Providers_Master[Column1] = "High Staffing")
 - OK Count = CALCULATE(COUNTROWS(Providers_Master),
Providers_Master[Column1] = "OK")
 - Providers per Specialty = DISTINCTCOUNT(Providers_Master[ProviderID])
 - Specialty Count per Provider = CALCULATE(COUNTROWS(Providers_Master),
ALLEXCEPT(Providers_Master, Providers_Master[Specialty]))
 - Top Specialty = TOPN(1, VALUES(Providers_Master[Specialty]), [Providers per Specialty])
 - Total Providers = COUNTROWS(Providers_Master)
-

Table 10: Vitals_Log Table Calculated Columns & DAX Measures

- % Stable = `DIVIDE(CALCULATE(COUNTROWS(Vitals_Log), Vitals_Log[Risk_Status]="Stable"), [Total Vitals])`
 - Avg Systolic = `AVERAGE(Vitals_Log[SystolicBP])`
 - BP Category = `SWITCH(TRUE(),
Vitals_Log[SystolicBP] < 120 && Vitals_Log[DiastolicBP] < 80, "Normal",
Vitals_Log[SystolicBP] < 130 && Vitals_Log[DiastolicBP] < 80, "Elevated",
Vitals_Log[SystolicBP] < 140 || Vitals_Log[DiastolicBP] < 90, "Hypertension Stage 1",
"Hypertension Stage 2")`
 - Distinct Patients = `DISTINCTCOUNT(Vitals_Log[Patient_ID])`
 - MAP = `Vitals_Log[DiastolicBP] + (Vitals_Log[SystolicBP] - Vitals_Log[DiastolicBP]) / 3`
 - Pulse Pressure = `Vitals_Log[SystolicBP] - Vitals_Log[DiastolicBP]`
 - Repeat Patients = `CALCULATE(DISTINCTCOUNT(Vitals_Log[Patient_ID]),
FILTER(VALUES(Vitals_Log[Patient_ID]),
CALCULATE(COUNTROWS(Vitals_Log)) > 1))`
 - Temperature_C = `ROUND((Temperature_F - 32) * 5/9, 1)`
 - Total Vitals = `COUNTROWS(Vitals_Log)`
-

8. Analysis and Visualizations (Power BI)

Dashboard Features:

Charges Distribution – KPI cards, donut/line/treemap/column visuals for charge distribution, diagnosis revenue and charge categories.

Diagnoses – Clinical Insights – KPI cards, ribbon/waterfall/pie/bar/table visuals for diagnosis prevalence, conditions and charge contribution.

Provider & Facility Performance – revenue/encounter KPIs, facility and provider visuals, specialty/charge analysis and slicers.

Claims + Revenue Cycle – billed, paid, remaining, collection-rate and claim-volume KPIs with insurer and payment visuals.

Patient Demographics Dashboard – patient count, age, gender, checkup routine, blood-group/age-band and demographic visuals.

Pharmacy & Revenue Overview – prescription revenue, medication, pharmacy and time-based analysis.

Pharmacy Performance – medication/pharmacy volume and revenue trends with funnel/treemap/area/column visuals.

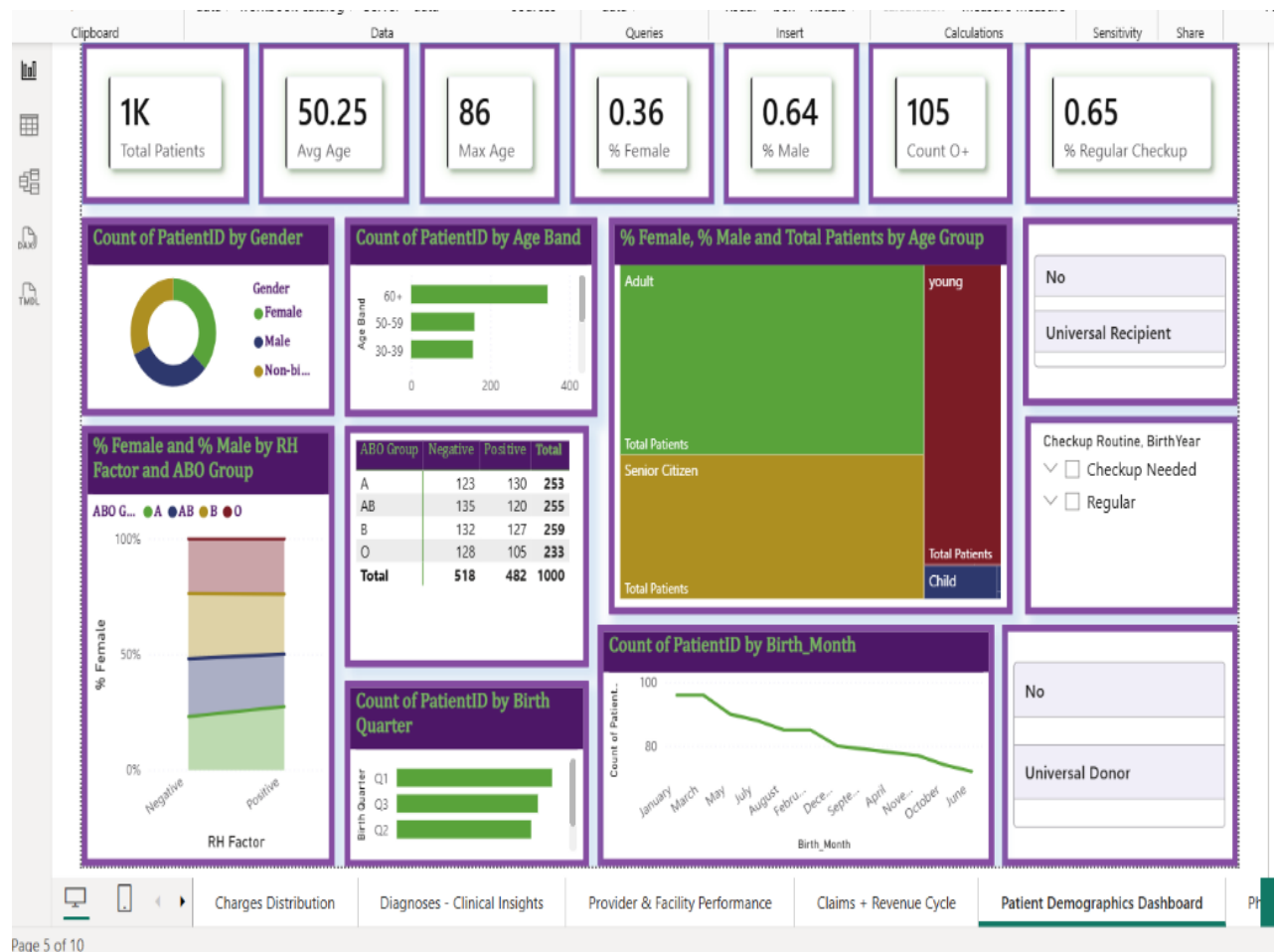
Procedure Volume Overview – procedure counts, minutes, duration and encounter-type analysis.

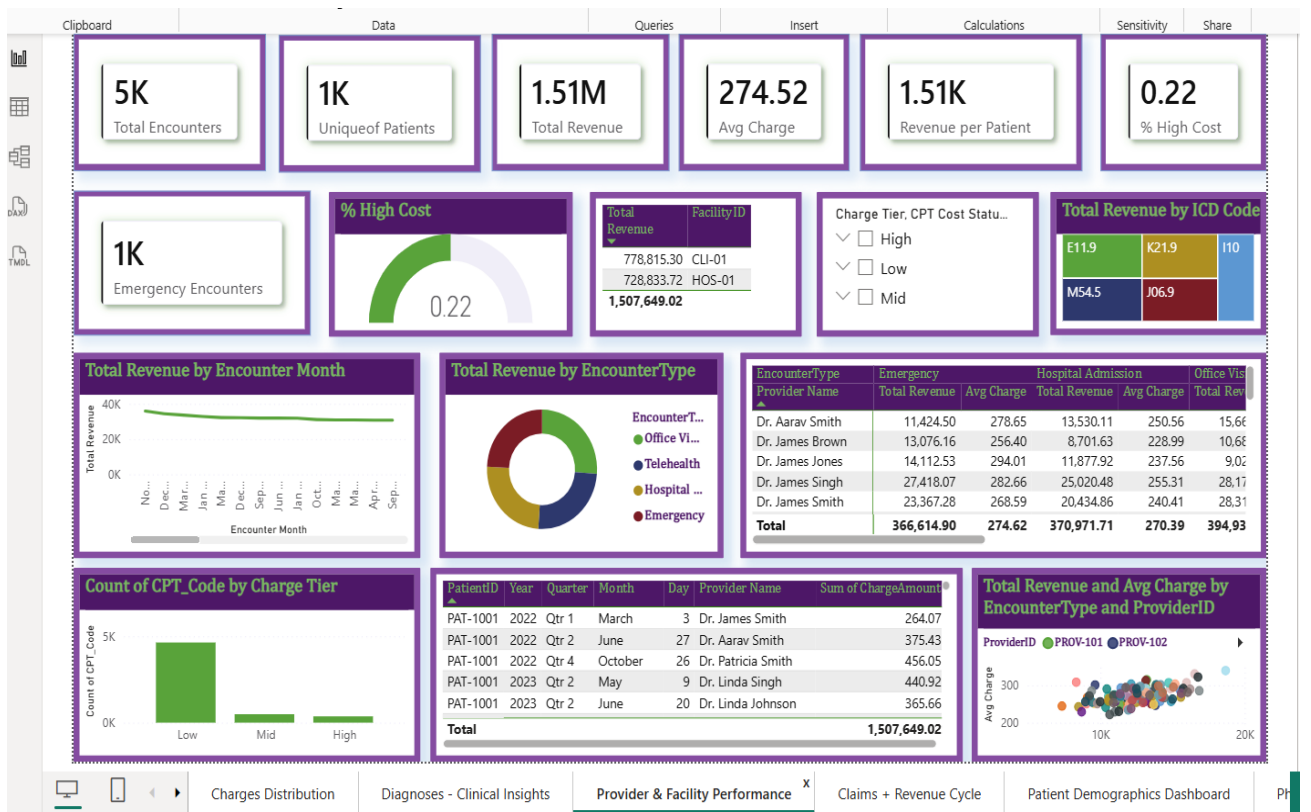
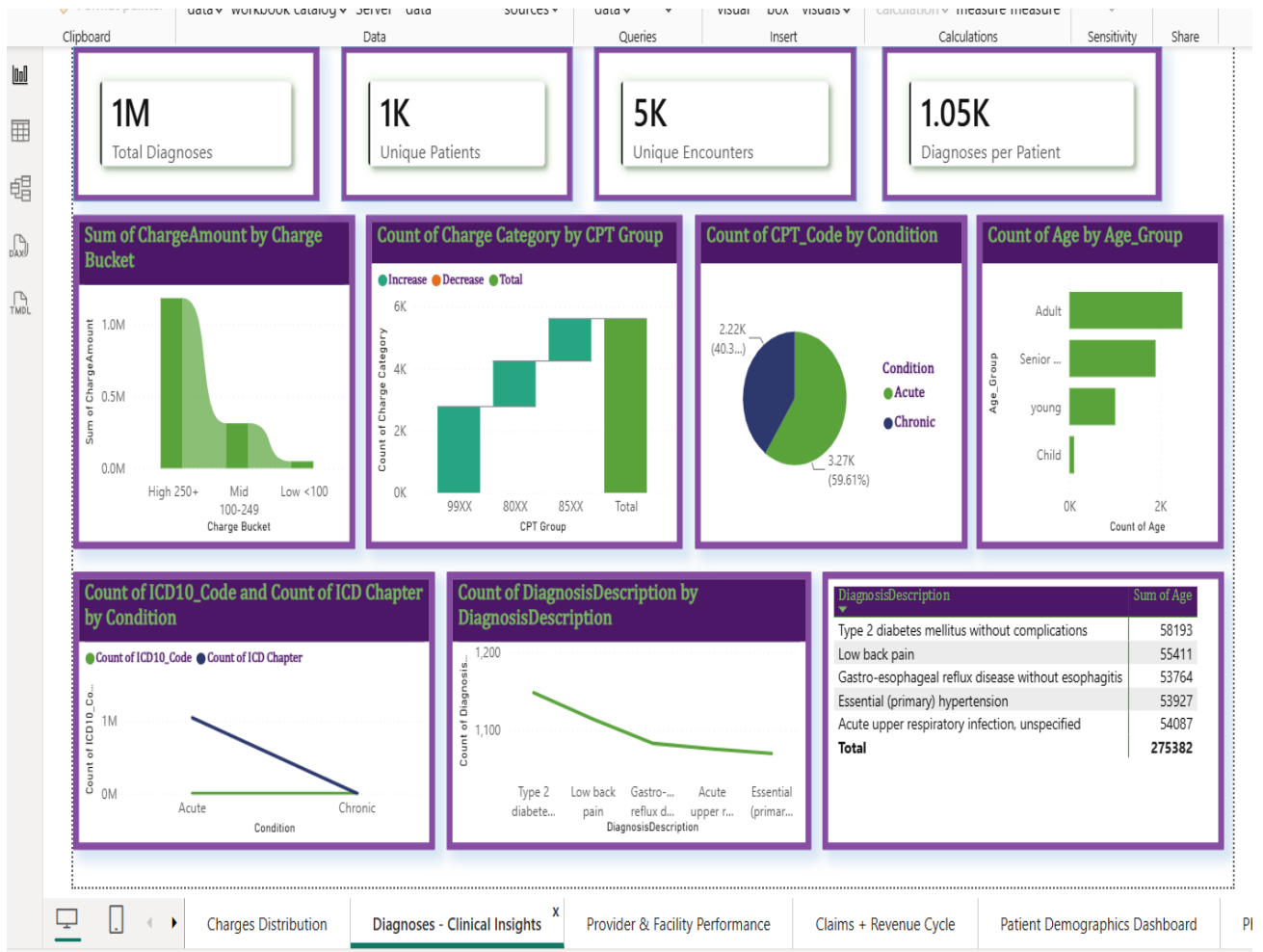
Provider Staffing Overview – provider count, staffing status, specialty distribution and provider tables.

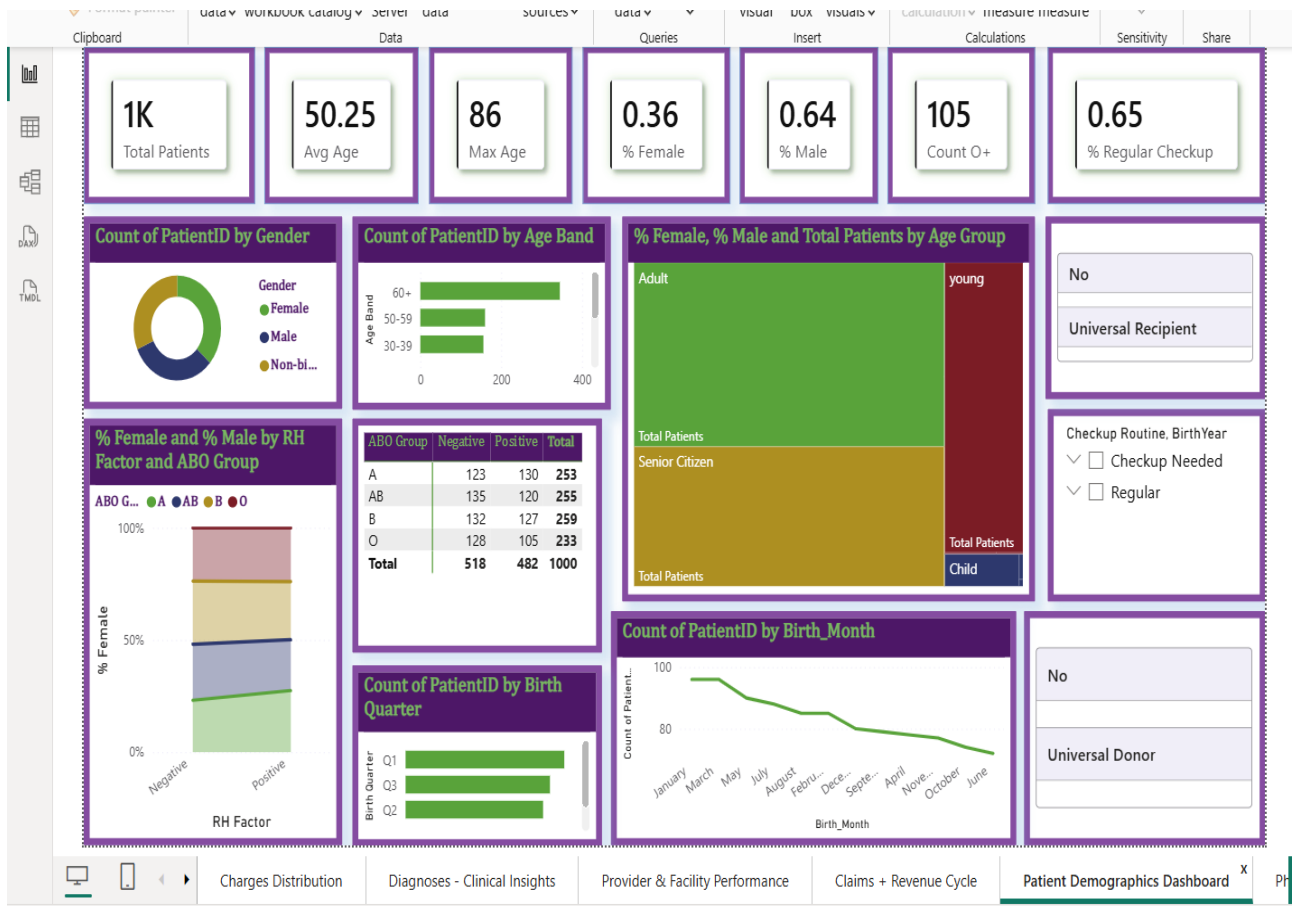
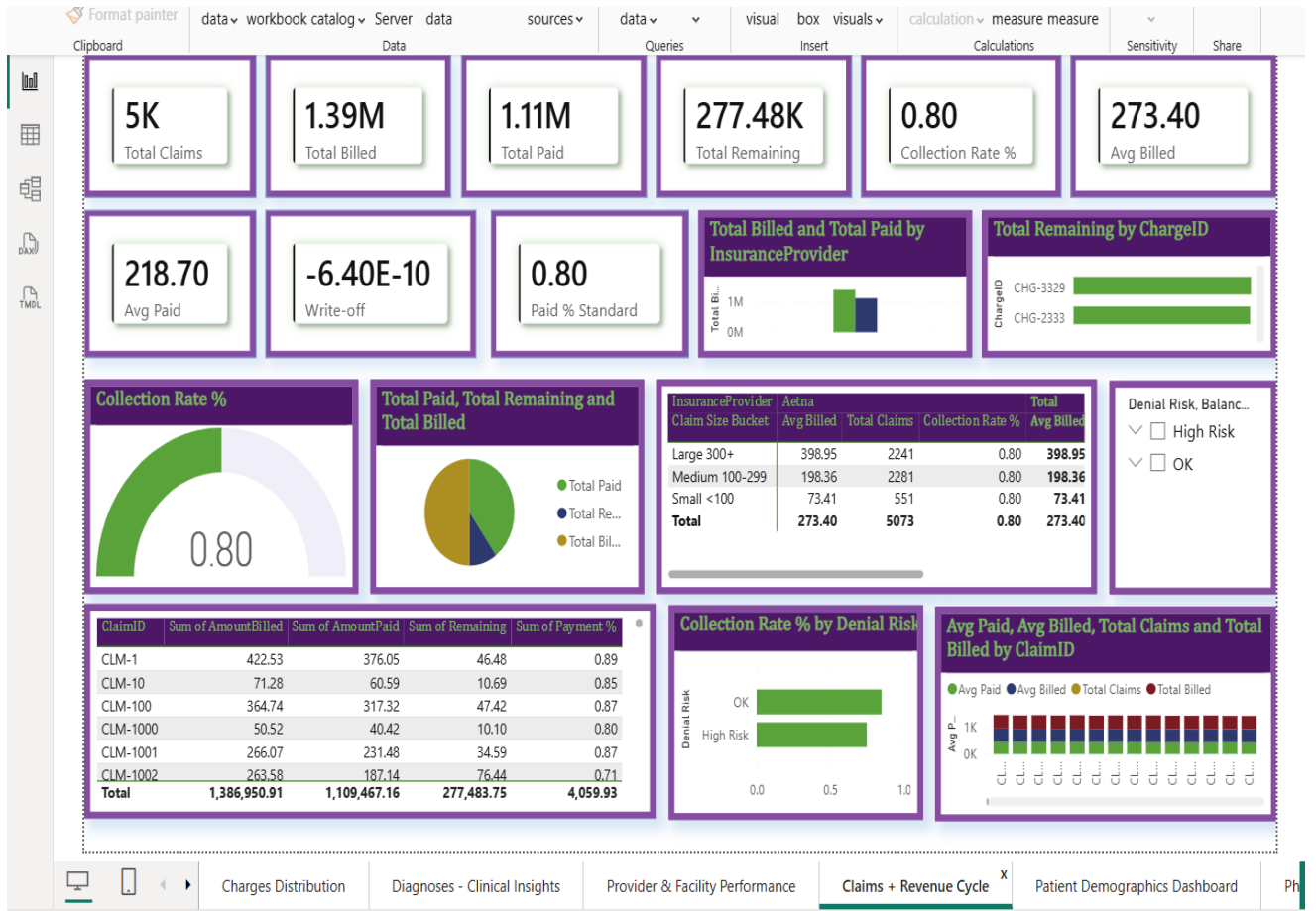
Vitals & Risk Overview – vitals count, patient count, systolic pressure, stable-risk rate and risk/category trends.

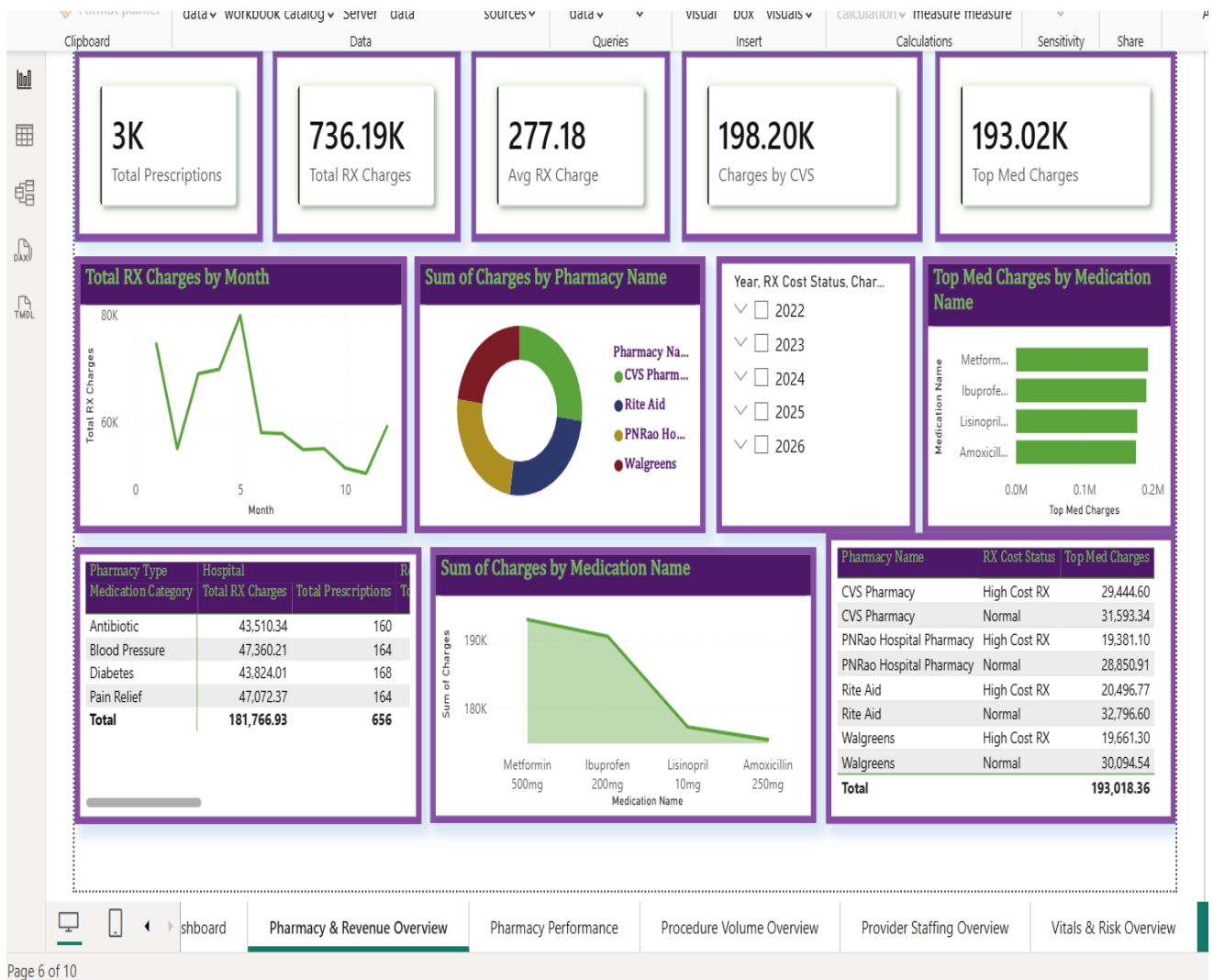
Interactivity: the PBIX layout includes slicers and drill/filter-capable visuals. Clear visual titles and KPI cards are used throughout the report; the page structure supports consolidated operational, clinical and revenue-cycle reporting.

SCREENSHOT OF DASHBOARD /REPORT





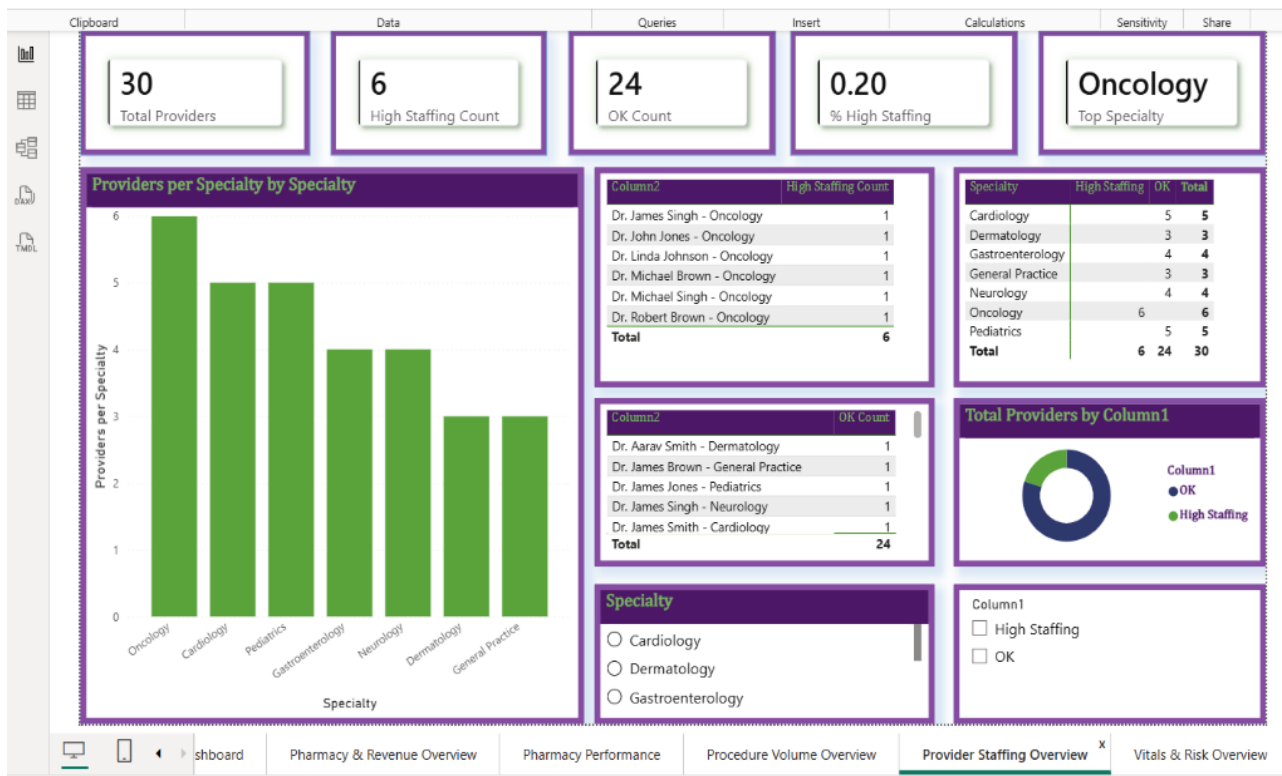
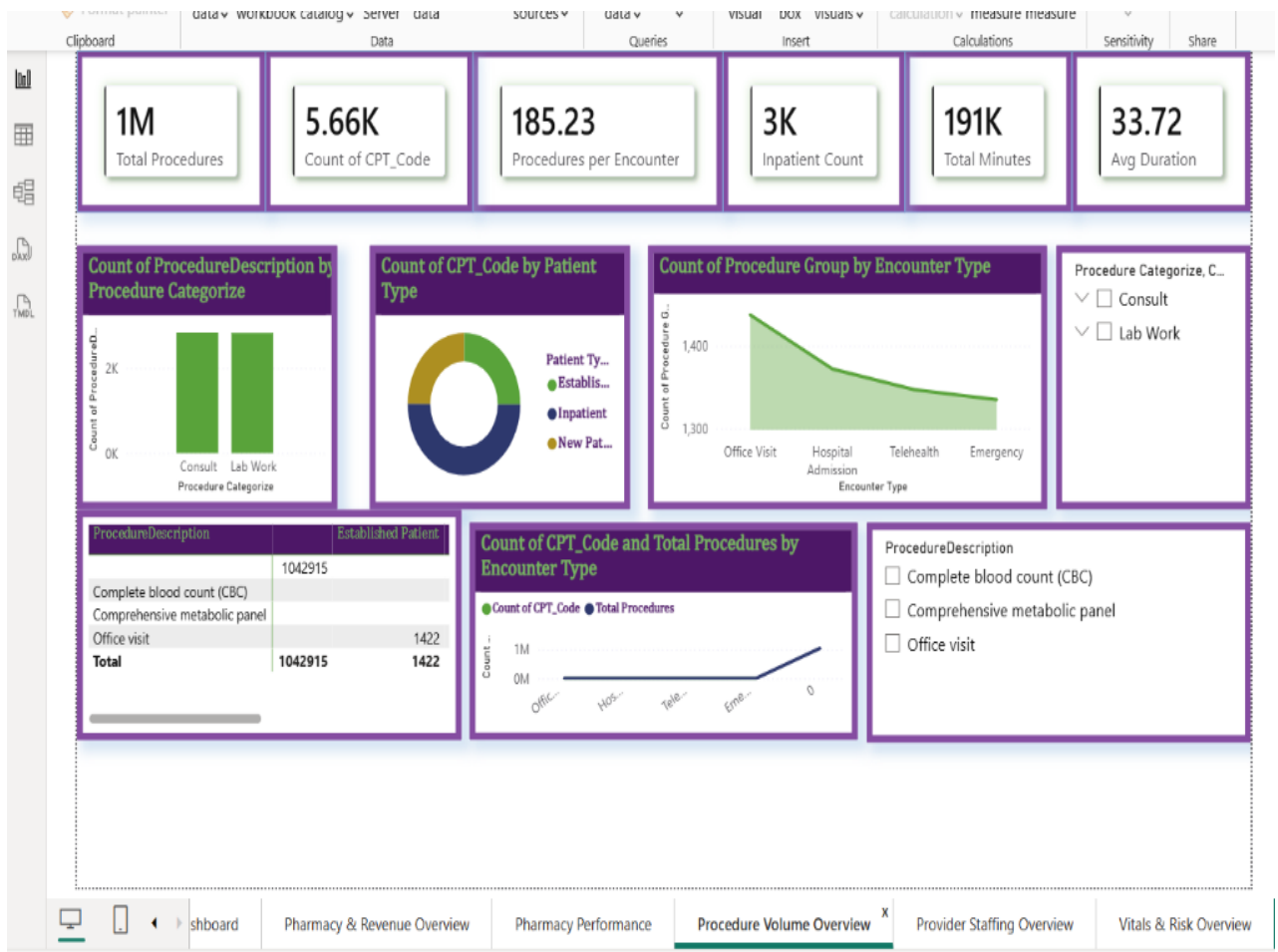


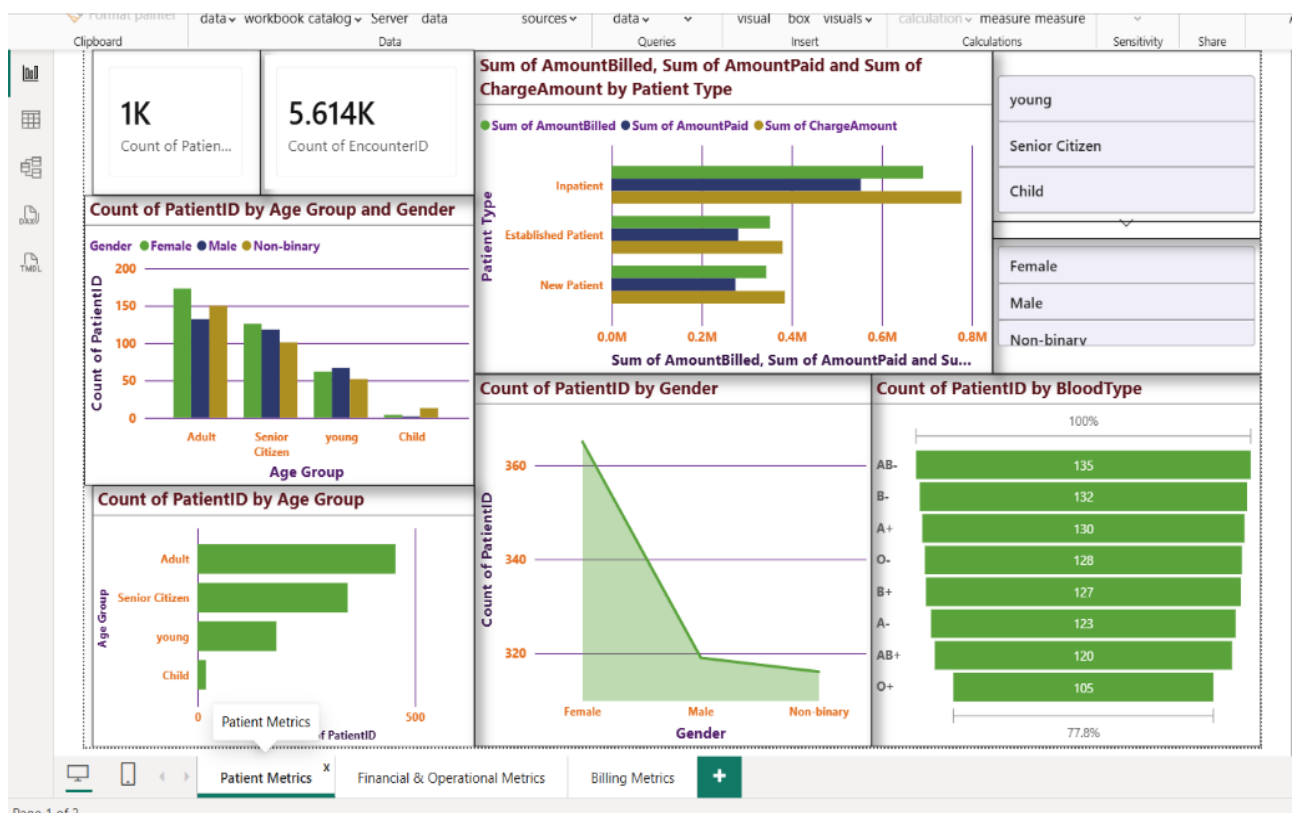


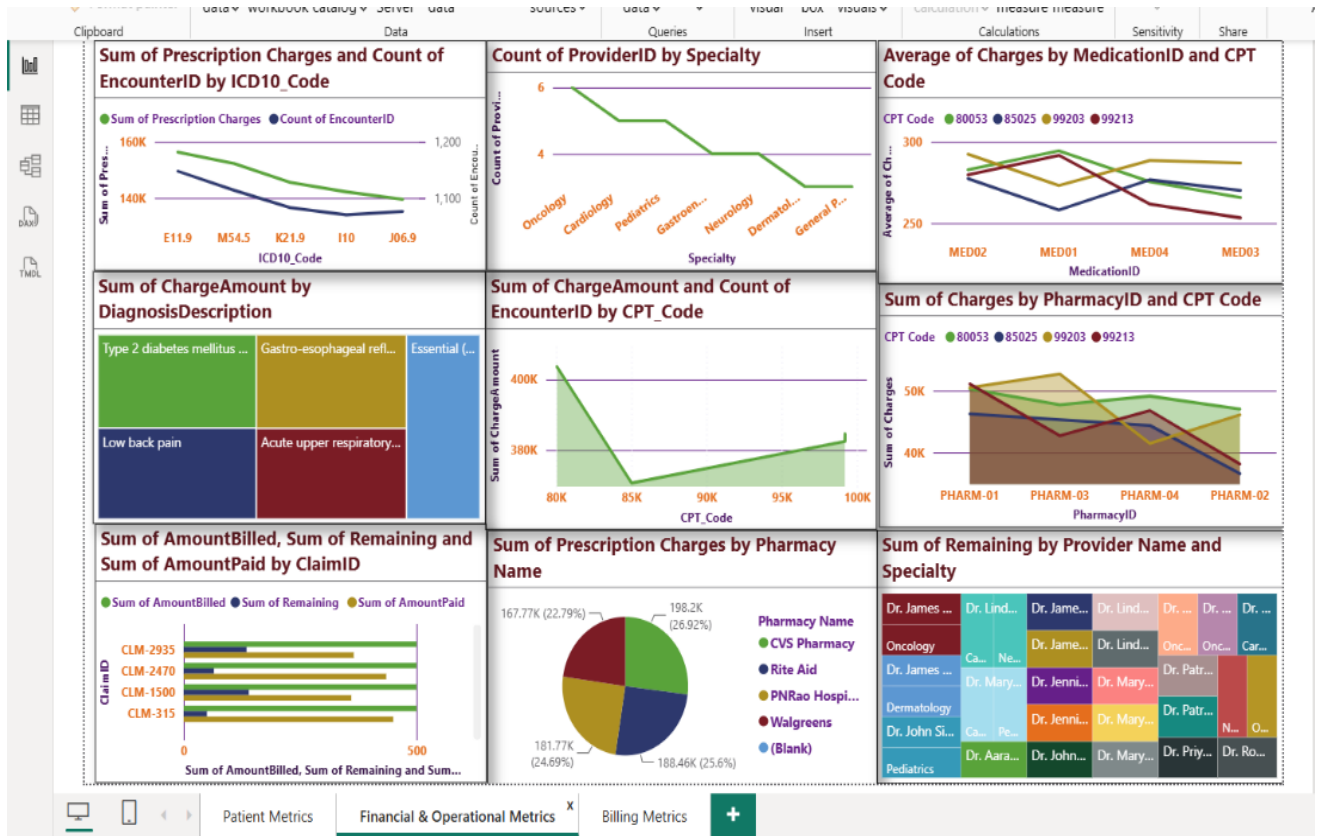
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Report view

- Cardiology
- Dermatology
- Gastroenterology
- CVS Pharmacy
- PNRao Hospital Pharmacy
- Rite Aid

Provider Name	Specialty	Patient Type	PatientID
Dr. Aarav Smith	Dermatology	Inpatient	PAT-1001
Dr. James Brown	General Practice	Inpatient	PAT-1001
Dr. James Smith	Dermatology	Inpatient	PAT-1001
Dr. Linda Johnson	Oncology	Inpatient	PAT-1001
Dr. Linda Singh	Neurology	New Patient	PAT-1001
Dr. Mary Garcia	General Practice	Inpatient	PAT-1001
Dr. Michael Singh	Oncology	New Patient	PAT-1001
Dr. Patricia Smith	Pediatrics	Inpatient	PAT-1001
Dr. Robert Brown	Oncology	New Patient	PAT-1001
Dr. Aarav Smith	Dermatology	Inpatient	PAT-1002
Dr. James Brown	General Practice	Established Patient	PAT-1002

Pharmacy Name	Prescription Charges	Month	Medication Name	ICD10_Code	PatientID
Walgreens	315.19	January	Amoxicillin 250mg	E11.9	PAT-1329
Walgreens	175.08	January	Amoxicillin 250mg	I10	PAT-1401
Walgreens	424.54	January	Amoxicillin 250mg	I10	PAT-1490
Walgreens	231.15	January	Amoxicillin 250mg	I10	PAT-1514
Walgreens	246.55	January	Amoxicillin 250mg	J06.9	PAT-1019
Walgreens	415.47	January	Amoxicillin 250mg	K21.9	PAT-1341
Total	736,194.88				

FullName	80053	85025	99203	99213	Total
Dr. Aarav Smith	12,552.77	11,972.78	11,197.95	14,414.49	50,137.99
Dr. James Brown	9,712.19	12,824.98	13,420.44	11,670.48	47,628.09
Dr. James Jones	11,119.34	8,556.92	10,852.14	12,400.14	42,928.54
Dr. James Singh	26,207.78	30,698.91	20,264.14	21,645.54	98,816.37
Dr. James Smith	20,774.65	26,498.17	19,329.09	26,606.43	93,208.34
Dr. Jennifer Garcia	9,773.41	8,465.78	10,664.34	11,912.52	40,816.05
Dr. Jennifer Williams	10,881.28	10,358.91	14,035.96	10,631.42	45,907.57
Dr. John Jones	5,112.91	11,326.26	13,077.71	9,829.14	39,346.02
Dr. John Singh	23,998.46	21,712.35	24,877.50	21,142.05	91,730.36
Dr. Linda Johnson	11,247.70	11,154.93	11,943.10	11,054.19	45,399.92
Dr. Linda Singh	23,327.78	24,181.85	24,374.33	29,703.82	101,587.78
Dr. Linda Smith	13,432.64	11,466.62	10,081.09	9,537.48	44,517.83
Dr. Mary Brown	11,239.87	12,440.59	13,133.12	9,258.63	46,072.21
Total	334,281.50	357,681.79	343,424.43	351,563.19	1,386,950.91

PatientID	Year	AmountBilled	AmountPaid	Remaining	CPT_Code
PAT-1001		331.74	265.39	66.35	80053
PAT-1001		686.19	610.71	75.48	85025
PAT-1001		801.72	686.51	115.21	99203
PAT-1001	2022	346.53	291.09	55.44	85025
PAT-1001	2024	396.93	321.51	75.42	85025
PAT-1001	2025	463.75	338.54	125.21	85025
PAT-1002		71.28	60.59	10.69	80053
PAT-1002		347.49	302.32	45.17	99203
PAT-1002		394.06	279.78	114.28	99213
PAT-1002	2022	461.37	372.27	89.10	99213
PAT-1002	2025	121.36	104.37	16.99	85025
PAT-1002	2026	498.93	444.05	54.88	80053
PAT-1003		468.47	332.61	135.86	85025
PAT-1003	2022	379.02	265.31	113.71	80053
Total		1,386,950.91	1,109,467.16	277,483.75	

Patient Metrics | Financial & Operational Metrics | Billing Metrics

9. Insights & Conclusions

Provide the analysis insights:

Descriptive

- The Excel dataset contains 1,000 patient master records, with an average patient age of 50.3 years.
- Gender distribution is Female 365, Male 319, and Non-binary 316.
- The analysed encounter/charge data contains 5,491 encounters and approximately \$1,507,649.02 in encounter charges.
- Insurance claims contain 5,073 claims, with \$1,386,950.91 billed, \$1,109,467.16 paid and \$277,483.75 remaining.
- Prescription activity contains 2,656 prescriptions with \$736,194.88 in prescription charges.
- Procedure activity contains 5,660 procedure records and 5,614 vital-sign records.

Diagnosis

- Office Visits are the highest-revenue encounter category at approximately \$394,938.27; Telehealth and Hospital Admission also contribute substantial revenue.
- CPT 80053 is the highest-revenue CPT code in the Excel charge summary at approximately \$391,616.27.
- The highest-revenue diagnosis code in the encounter data is E11.9, contributing approximately \$316,422.20.
- The highest-revenue provider in the analysed provider-revenue view is Dr. James Singh at approximately \$111,400.46.
- On the insurance side, the supplied Excel summary is entirely associated with Aetna, so payer comparison is not possible without additional insurer records.

Predictive

- Historical monthly encounter, charge and prescription trends in Power BI can be used as a baseline for forecasting demand and revenue.
- Repeated vitals and diagnosis patterns can be used to flag patient cohorts that may require closer monitoring.
- High-cost encounter/service categories can be monitored to anticipate resource and revenue-cycle pressure.

Prescriptive

- Prioritise high-revenue encounter and CPT categories for capacity and coding-quality review.
- Investigate outstanding insurance balances and payment leakage to improve collection performance.
- Use provider/specialty revenue and staffing indicators to align workforce capacity with demand.
- Use pharmacy-level and medication-level revenue patterns to optimise inventory, formulary planning and prescription workflows.
- Use demographic and clinical risk segmentation to support targeted preventive-care and follow-up programmes.

10. Conclusions

The integration of Excel and Power BI provides an effective end-to-end healthcare analytics workflow, from structured data preparation and PivotTable validation through DAX-based modelling and interactive dashboard reporting. The project demonstrates how patient, clinical, operational and financial data can be consolidated into a single decision-support framework. The resulting Power BI report supports descriptive, diagnostic, predictive and prescriptive analysis, while the Excel workbook provides transparent source-level calculations and validation summaries.